The Laplace Transform

1 The Laplace Transform

Definition 1.1 (The Laplace Transform)**.**

The (bilateral) Laplace transform of a general signal $x(t)$ is defined as

$$X(s) = \int_{-\infty}^{\infty} x(t)\mathrm{e}^{-st}\,\mathrm{d}t$$

where $s = \sigma + \mathrm{j}\omega$ is a complex variable.

Remark. **Connection with the Fourier Transform**

$$\mathcal{L}\{x(t)\} = X(s) = X(\sigma + \mathrm{j}\omega) = \mathcal{F}\{x(t)\mathrm{e}^{-\sigma t}\}$$

Definition 1.2 (The Inverse Laplace Transform)**.**

The inverse Laplace transform of a function $X(s)$ is defined as

$$x(t) = \frac{1}{2\pi\mathrm{j}} \int_{\sigma-\mathrm{j}\infty}^{\sigma+\mathrm{j}\infty} X(s)\mathrm{e}^{st}\,\mathrm{d}s$$

***Proof*.**

Use the inverse Fourier transform:

$$x(t)\mathrm{e}^{-\sigma t} = \mathcal{F}^{-1}\{X(\sigma + \mathrm{j}\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\sigma + \mathrm{j}\omega)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega$$

□

2 Properties of the ROC

1. The ROC consists of strips parallel to the $\mathrm{j}\omega$-axis in the s-plane.
2. The ROC dose not include any poles.
3. If $x(t)$ is of finite duration and is absolutely integrable, then the ROC is the entire s-plane.
4. If $x(t)$ is right-sided, then the ROC is of the form $\mathcal{R}e(s) > \sigma_0$ for some σ_0.
5. If $x(t)$ is left-sided, then the ROC is of the form $\mathcal{R}e(s) < \sigma_0$ for some σ_0.
6. If $x(t)$ is two-sided, then the ROC is of the form $\sigma_1 < \mathcal{R}e(s) < \sigma_2$ for some σ_1 and σ_2.

3 Basic Laplace Transform Pairs

Signal	Laplace Transform	ROC
$u(t)$	$\frac{1}{s}$	$\mathcal{R}e\{s\} > 0$
$-u(-t)$	$\frac{1}{s}$	$\mathcal{R}e\{s\} < 0$
$\mathrm{e}^{-b\|t\|}$	$\frac{-2b}{s^2 - b^2}$	$-\mathcal{R}e\{b\} < \mathcal{R}e\{s\} < \mathcal{R}e\{b\}$
$\mathrm{e}^{-at}u(t)$	$\frac{1}{s+a}$	$\mathcal{R}e\{s+a\} > 0$
$-\mathrm{e}^{-at}u(-t)$	$\frac{1}{s+a}$	$\mathcal{R}e\{s+a\} < 0$
$\frac{t^{n-1}}{(n-1)!}\mathrm{e}^{-at}u(t)$	$\frac{1}{(s+a)^n}$	$\mathcal{R}e\{s+a\} > 0$
$-\frac{t^{n-1}}{(n-1)!}\mathrm{e}^{-at}u(-t)$	$\frac{1}{(s+a)^n}$	$\mathcal{R}e\{s+a\} < 0$
$\delta(t)$	1	$\mathbb{C}$
$\delta(t-T)$	e^{-sT}	$\mathbb{C}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$\mathcal{R}e\{s\} > 0$
$\cos(\omega_0 t)u(t)$	$\frac{s}{s^2+\omega_0^2}$	$\mathcal{R}e\{s\} > 0$
$\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$	$\mathcal{R}e\{s\} > 0$
$\mathrm{e}^{-at}\cos(\omega_0 t)u(t)$	$\frac{s+a}{(s+a)^2+\omega_0^2}$	$\mathcal{R}e\{s+a\} > 0$
$\mathrm{e}^{-at}\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{(s+a)^2+\omega_0^2}$	$\mathcal{R}e\{s+a\} > 0$
$u_n(t) = \frac{\mathrm{d}^n\delta(t)}{\mathrm{d}t^n}n$	s^n	$\mathbb{C}$
$u_{-n}(t) = \underbrace{u(t) * \cdots * u(t)}_{n \text{ times}}$	$\frac{1}{s^n}$	$\mathcal{R}e\{s\} > 0$

4 Properties of the Bilateral Laplace Transform

4.1

Signal	Transform	ROC
$x(t)$	$X(s)$	R
$x_1(t)$	$X_1(s)$	R_1
$x_2(t)$	$X_2(s)$	R_2
$ax_1(t) + bx_2(t)$	$aX_1(s) + bX_2(s)$	At least $R_1 \cap R_2$
$x(t - t_0)$	$\mathrm{e}^{-st_0}X(s)$	R
$\mathrm{e}^{s_0 t}x(t)$	$X(s - s_0)$	$R + \mathcal{R}e\{s_0\}$
$x(at)$	$\frac{1}{\|a\|}X\left(\frac{s}{a}\right)$	aR
$x^*(t)$	$X^*(s^*)$	R
$x_1(t) * x_2(t)$	$X_1(s)X_2(s)$	At least $R_1 \cap R_2$
$\frac{\mathrm{d}}{\mathrm{d}t}x(t)$	$sX(s)$	At least R
$-tx(t)$	$\frac{\mathrm{d}}{\mathrm{d}s}X(s)$	R
$\int_{-\infty}^{t} x(\tau)\,\mathrm{d}\tau$	$\frac{1}{s}X(s)$	At least $R \cap \{\mathcal{R}e\{s\} > 0\}$

4.2 Initial- and Final-Value Theorem

Theorem 4.1 (Initial-Value and Final-Value Theorem)**.**

If $x(t) = 0$ for $t < 0$ and $x(t)$ contains no impulses or higher-order singularities at $t = 0$, then

$$x(0^+) = \lim_{s\to+\infty} sX(s)$$
$$\lim_{t\to+\infty} x(t) = \lim_{s\to 0} sX(s)$$

5 The Unilateral Laplace Transform

5.1

Definition 5.1 (The Unilateral Laplace Transform)**.**

The unilateral Laplace transform of a general signal $x(t)$ is defined as

$$\mathfrak{X}(s) = \int_{0^-}^{\infty} x(t)\mathrm{e}^{-st}\,\mathrm{d}t$$

where $s = \sigma + \mathrm{j}\omega$ is a complex variable.

5.2 Properties of the Unilateral Laplace Transform

For causal signals, the unilateral Laplace transform is the same as the bilateral Laplace transform.

5.2.1

Signal	Transform
$x(t)$	$\mathfrak{X}(s)$
$x_1(t)$	$\mathfrak{X}_1(s)$
$x_2(t)$	$\mathfrak{X}_2(s)$
$ax_1(t) + bx_2(t)$	$a\mathfrak{X}_1(s) + b\mathfrak{X}_2(s)$
$\mathrm{e}^{s_0 t} x(t)$	$\mathfrak{X}(s - s_0)$
$x(at) \qquad a > 0$	$\frac{1}{a}\mathfrak{X}\left(\frac{s}{a}\right)$
$x^*(t)$	$\mathfrak{X}^*(s^*)$
$x_1(t) * x_2(t)$ both causal	$\mathfrak{X}_1(s)\mathfrak{X}_2(s)$
$\frac{\mathrm{d}}{\mathrm{d}t}x(t)$	$s\mathfrak{X}(s) - x(0^-)$
$-tx(t)$	$\frac{\mathrm{d}}{\mathrm{d}s}\mathfrak{X}(s)$
$\int_{0^-}^{t} x(\tau)\,\mathrm{d}\tau$	$\frac{1}{s}\mathfrak{X}(s)$

5.2.2 Initial- and Final-Value Theorem

Theorem 5.1 (Initial-Value and Final-Value Theorem)**.**

For any $x(t)$*,*

$$x(0^+) = \lim_{s \to +\infty} s\mathfrak{X}(s)$$

$$\lim_{t \to +\infty} x(t) = \lim_{s \to 0} s\mathfrak{X}(s)$$